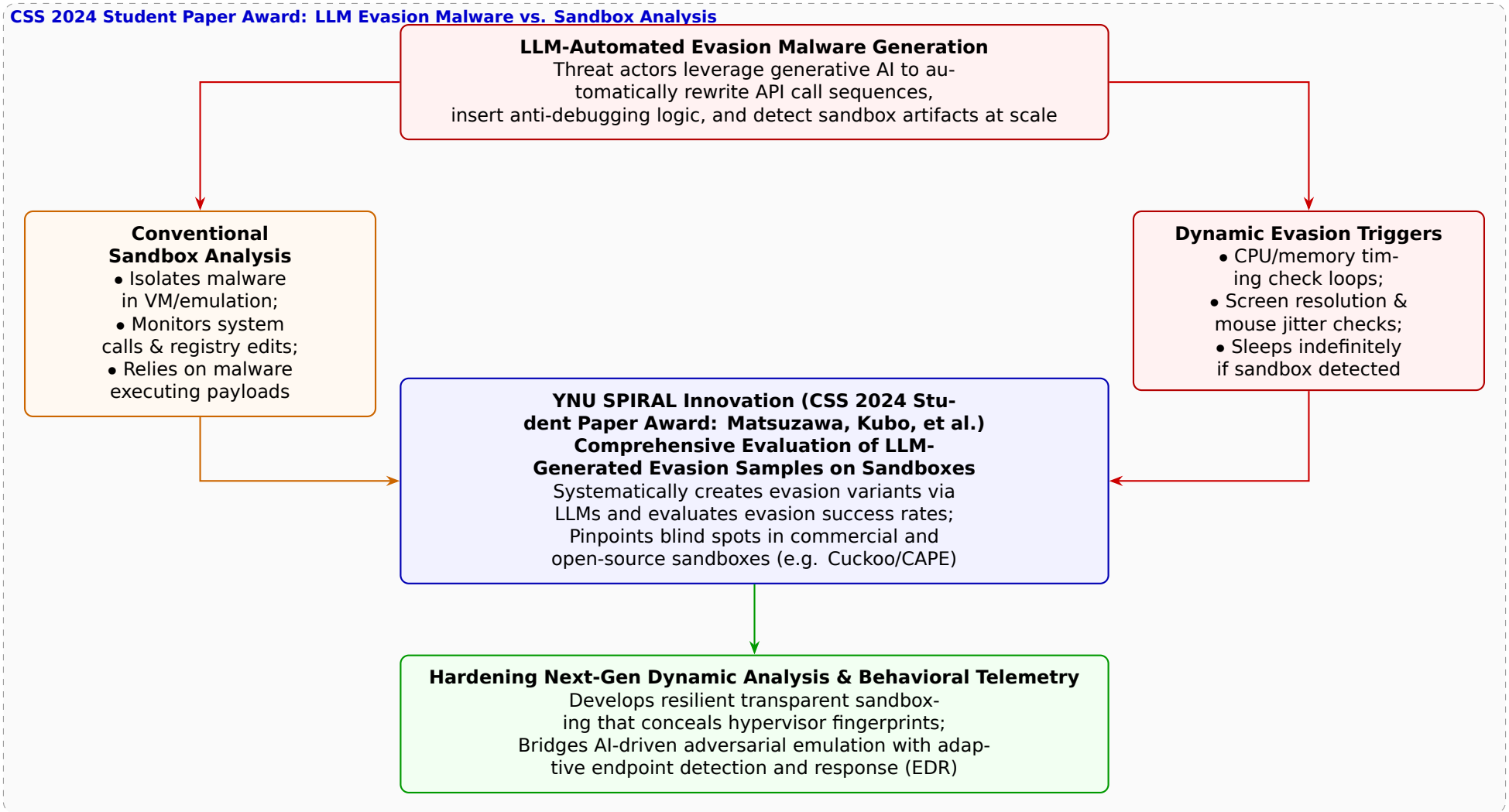
CSS 2024 Student Paper Award: LLM Evasion Malware vs. Sandbox Analysis
LLM-Automated Evasion Malware Generation
Threat actors leverage generative AI to automatically rewrite API call sequences, insert anti-debugging logic, and detect sandbox artifacts at scale
Conventional Sandbox Analysis
• Isolates malware in VM/emulation;
• Monitors system calls & registry edits;
• Relies on malware executing payloads
Dynamic Evasion Triggers
• CPU/memory timing check loops;
• Screen resolution & mouse jitter checks;
• Sleeps indefinitely if sandbox detected
YNU SPIRAL Innovation (CSS 2024 Student Paper Award: Matsuzawa, Kubo, et al.)
Comprehensive Evaluation of LLM-Generated Evasion Samples on Sandboxes
Systematically creates evasion variants via LLMs and evaluates evasion success rates;
Pinpoints blind spots in commercial and open-source sandboxes (e.g. Cuckoo/CAPE)
Hardening Next-Gen Dynamic Analysis & Behavioral Telemetry
Develops resilient transparent sandboxing that conceals hypervisor fingerprints;
Bridges AI-driven adversarial emulation with adaptive endpoint detection and response (EDR)